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• Question #1:-

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$A = \{2, 4, 6\} \quad B = \{1, 2, 3, 4\}$$

• $A \cup B = \{2, 4, 6\} \cup \{1, 2, 3, 4\} = \{1, 2, 3, 4, 6\}$

• $A \cap B = \{2, 4, 6\} \cap \{1, 2, 4, 3\} = \{2, 4\}$

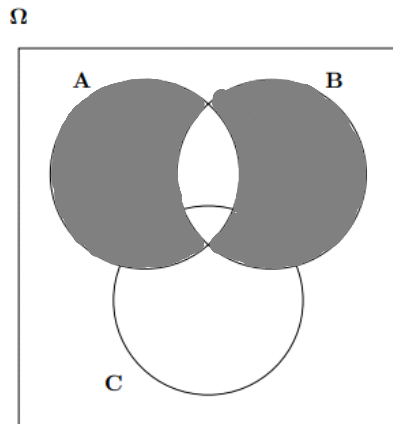
• $A' = \{1, 3, 5\}$

• Question #2

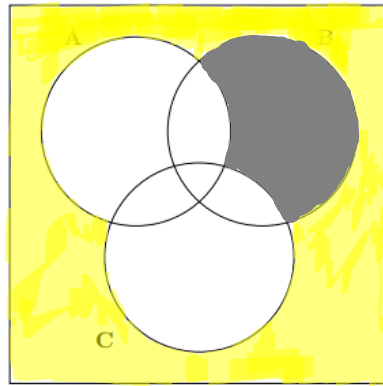
• $A = \{13 \text{ cards of hearts}\}, B = \{12 \text{ face cards}\}, C = \{4 \text{ ace cards}\}$

• Using "yellow" color to colour superset to distinguish amongst sets.

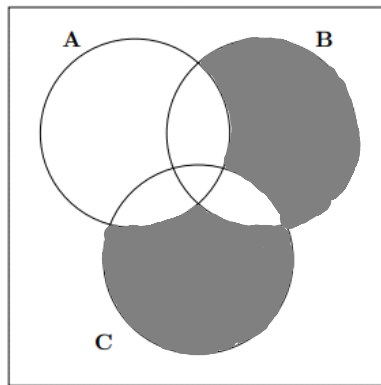
1. The event that the card is either a heart or a face card but not both.



2. The event that the card is neither a heart nor an ace.

Ω 

3. The event that the card is either an ace or a face card, but not both, and also not a heart.

 Ω 

• Question #3:-

Sample space = {Red, green, blue}

Total possibilities = 3 red balls + 2 blue balls + 1 green balls
= 6 balls

- probability of getting a red ball = $\frac{3}{6} = \frac{1}{2} = 0.5$
- probability of getting a green ball = $\frac{1}{6} = 0.166$
- probability of getting a blue ball = $\frac{2}{6} = \frac{1}{3} = 0.333$

- probability of getting a non-red ball = $\frac{1}{3} + \frac{1}{6} = \frac{3}{6} = \frac{1}{2} = 0.5$

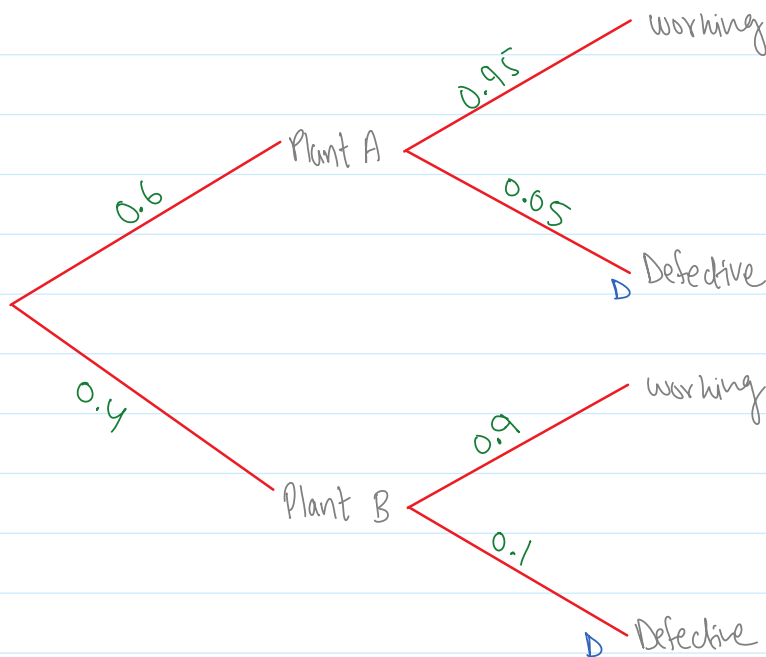
• Question # 4:-

A factory produces 60% of its products in Plant A and 40% in Plant B.

- 5% of Plant A's products are defective.
- 10% of Plant B's products are defective.

If a product is chosen at random and found to be defective, what is the probability that it came from Plant B?

- let us consider the following tree:-



we want $P(B/D) = \frac{P(B \cap D)}{P(D)}$

$$P(D) = P(B \cap D) + P(A \cap D)$$

$$P(B \cap D) = 0.1 \times 0.4 = 0.04$$

$$P(A \cap D) = 0.05 \times 0.6 = 0.03$$

$$P(D) = 0.04 + 0.03 = 0.07$$

Now,

$$P(B|D) = \frac{0.04}{0.07} = 0.571$$

- There is a 57% chance that the detected piece came from plant 'B'.

• Question #5:-

A medical test is designed to detect a disease that occurs in 1% of the population.

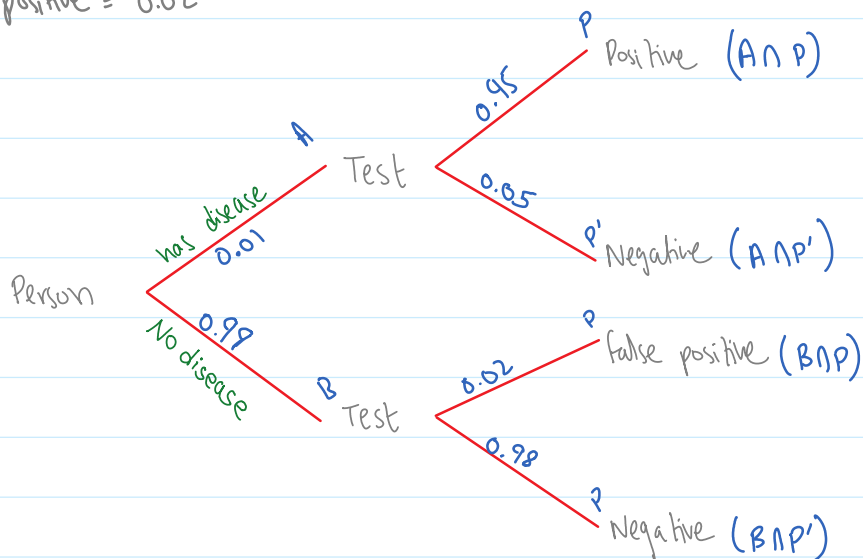
- If a person has the disease, the test is positive with probability 0.95.
- If a person does not have the disease, the test is (falsely) positive with probability 0.02.

A person is chosen at random and tests positive. What is the probability that the person actually has the disease?

Probability of disease = 0.01

test accuracy = 0.95

false positive = 0.02



$$P(A|P) = \frac{P(A \cap P)}{P(P)}$$

$$P(P) = P(A \cap P) + P(B \cap P) = 0.0095 + 0.0198 = 0.0293$$

$$\text{Probability of person actually positive} = \frac{0.0095}{0.0293} = 0.32$$

- The probability of person actually having the disease is 32%.

• Question # 6:-

Consider an experiment consisting of two successive rolls of a fair 4-sided die. Consider the following events in this experiment:

$A = \{\text{The result of first roll is less than 3}\}$

$B = \{\text{The result of second roll is greater than 2}\}$

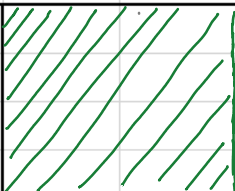
$C = \{\text{The sum of two rolls is greater than 6}\}$

Justify (by calculating appropriate probabilities) whether the following statements are true or false:

- (a) A and C are independent events.
- (b) A, B, and C are independent events.
- (c) A and B are not conditionally independent with respect to event C.

Consider the following table:-

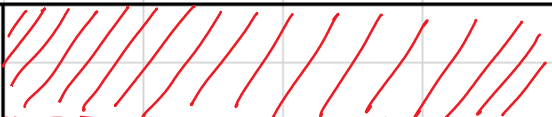
• for A:-

Second roll	4				
	3				
	2				
	1				
		1	2	3	4

first roll

$P(A) = 1/2$

• for B:-

Second roll	4				
	3				
	2				
	1				
		1	2	3	4

first roll

$P(B) = 1/2$

• for C:-

Second roll	4				
	3				
	2				
	1				

second roll

4				
3				
2				
1				
	1	2	3	4

first roll

$$P(C) = 3/16$$

- for this case $A \nmid C$ are simply not possible together so $P(ANC) = 0$

a) for A and C to be independent $P(ANC) = P(A) \times P(C)$

$$\text{so } P(ANC) = P(A) \times P(C)$$

$$= 1/2 \times 3/16 = 0.09 \neq P(ANC)$$

Hence $A \nmid C$ are not independent.

false

b) for $A, B \nmid C$ to be independent $P(ANBNC) = 0$,

false statement as A and B are not independent

Hence, A, B and C are not independent.

false

c) for $A \nmid B$ being conditionally independent. $(P(ANB)|C) = 0$

$$\text{So, } (P(ANB)|C) = P(A|C) \cdot P(B|C) = \frac{P(ANC)}{P(C)} \times \frac{P(BNC)}{P(C)}$$

since, $P(ANC) = 0$ so,

$$P(ANBNC) = 0$$

Now,

$$P(ANBNC|C) = \frac{0}{3/16} = 0$$

Now,

for R.H.S.:-

$$= \frac{0}{3/16} \times \frac{P(BNC)}{3/16} = 0$$

As, $P(A \cap B | C) = 0 = P(A|C) \cdot P(B|C)$
 The condition of independence is met

• The statement is false.

• Question #7:-

$$A = \{1, 2, 4\}$$

$$B = \{\text{same roll in both the dices}\}$$

second roll

6						
5						
4						
3						
2						
1						
	1	2	3	4	5	6

first roll

• Probability of $A = \frac{18}{36} = \frac{1}{2}$

• Probability of $B = \frac{6}{36} = \frac{1}{6}$

• for $P(A) \times P(B) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$

• We see that $P(A \cap B)$ from the table is $\frac{1}{36}$
 So,

$$P(A \cap B) = \frac{1}{12}$$

since $P(A \cap B) = P(A) \times P(B)$, so the events are independent.

• Question #8.-

- 4 digit number using $\{1, 2, 3, 4, 5\}$

$$\begin{array}{cccc} \underline{5} & \underline{5} & \underline{5} & \underline{5} \\ \text{1st} & \text{2nd} & \text{3rd} & \text{4th} \end{array} = 5^4 = 625$$

- Total number of possible digits = 625

• Question #9:-

- 1) $26^6 = 308915776$
 - There are 308915776 possible passwords that can be made by 26 upper case letters with repetition allowed.

- 2) there are 5 vowels and 21 consonants. so,
a password with no vowels:-

$$21 \times 21 \times 21 \times 21 \times 21 \times 21 = 85,766,121 \text{ possibilities}$$

Now,

$$\begin{aligned} \text{password containing at least 1 vowel} &= \text{Total} - \text{password with no vowels} \\ &= 308915776 - 85766121 = 223149655 \end{aligned}$$

- 223,149,655 possible passwords at least contain 1 vowel.

$$3) \frac{P(\text{contains vowel})}{P(\text{Total})} = \frac{223,149,655}{308,915,776} = 0.72$$

- There is a 72% chance that a randomly chosen password will contain at least one vowel.

• Question #10:

- 8 professors, 6 students = 14 people

i) at least 2-student means we can have 2, 3, 4, 5 students.

So,

- Case I: 2 students and 3 professors.

$$\text{ways to choose 2 students} = {}^6C_2 = 15$$

$$\text{ways to choose 3 professors} = {}^8C_3 = 56$$

$$\text{Total ways} = 56 \times 15 = 840$$

- Case II: 3 students and 2 professors

$$\text{ways for students} = \binom{6}{3} = 20$$

$$\text{ways for professors} = \binom{8}{2} = 28$$

$$\text{Total ways} = 20 \times 28 = 560$$

- Case III: 4 students 1 professor.

$$= \binom{6}{4} = 15$$

$$= \binom{8}{1} = 8$$

$$\text{Total ways} = 15 \times 8 = 120$$

- Case IV: 5 students 0 professor

$$\text{Students} = \binom{6}{5} = 6$$

$$\text{Professors} = \binom{8}{0} = 1$$

$$\text{Total ways} = 6 \times 1 = 6$$

Total ways in which a committee could be formed are:-

Case 1 + Case 2 + Case 3 + Case 4:-

$$840 + 560 + 120 + 6 = 1526 \text{ ways}$$

2) Randomly chosen committee:-

Total possibilities of a random committee = ${}^{19}C_5 = 2002$

Possibility of exactly 3 professors and 2 students = 840

So,

$$P(P_3, S_2) = \frac{840}{2002} = 0.419$$

- There is almost 42% chance that a randomly chosen committee will have exactly 3 professors and 2 students.

• Question # 11:-

- Total number of possibilities to choose 3-dots from 15:-

$${}^{15}C_3 = \binom{15}{3} = 455 \text{ ways.}$$

- if the dots make an equilateral triangle, then this means that they must have equal distance between them (ie. equal number of dots between them).

for that:-

$$15 \div 5$$

for that :-

$$15/3 = 5$$

To make equilateral triangle, let's suppose the dots are in three pairs on the circle, and each pair is numbered from (1,5).

So,

To form equal triangle following dots can be chosen.

$\{1, 2, 3, 4, 5\}$ $\{6, 7, 8, 9, 10\}$ $\{11, 12, 13, 14, 15\}$

- $\{1, 6, 11\}$, $\{2, 7, 12\}$, $\{3, 8, 13\}$, $\{4, 9, 14\}$, $\{5, 10, 15\}$

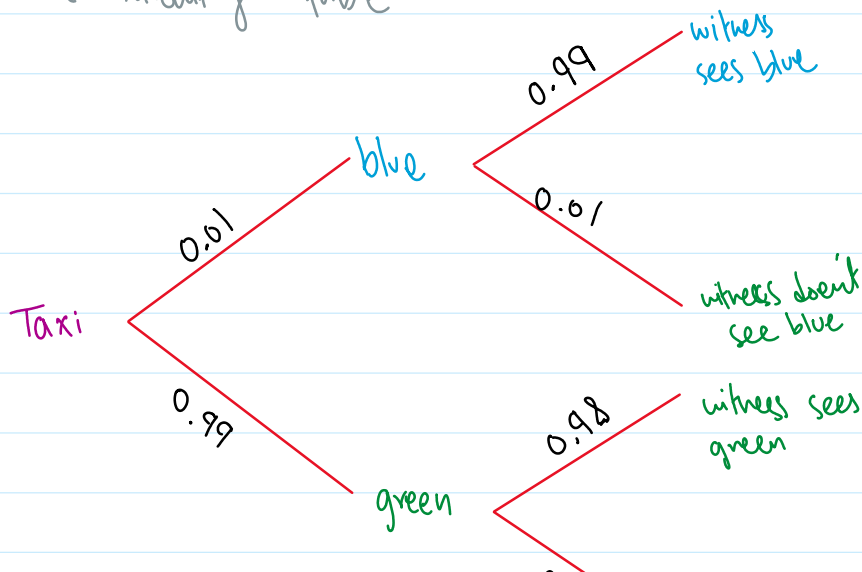
all of these points will form an equilateral triangle.

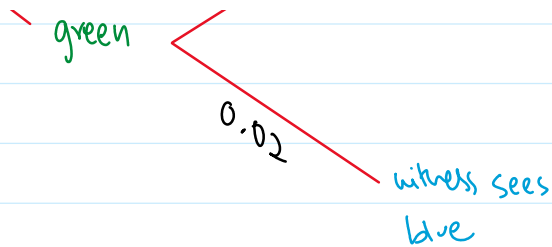
- number of dots that do not form equilateral triangle = Total - Those who form equilateral triangle
 $= 15 - 5 = 10$ pair of dots will not form equilateral triangle

• Question #12:-

In a city with one hundred taxis, 1 is blue and 99 are green. A witness observes a hit-and-run by a taxi at night and recalls that the taxi was blue, so the police arrest the blue taxi driver who was on duty that night. The driver proclaims his innocence and hires you to defend him in court. You hire a scientist to test the witness's ability to distinguish blue and green taxis under conditions similar to the night of accident. The data suggests that the witness sees blue cars as blue 99% of the time and green cars as blue 2% of the time. What is the probability that taxi was actually blue given the witness's statement?

consider the following table





- Total probability that witness says the car is blue:-
$$P(w) = 0.02 \times 0.99 + 0.01 \times 0.99 = 0.0198 + 0.0099 = 0.0297$$
- The actual probability of car being blue = $0.01 \times 0.99 = 0.0099$
- Probability of car actually being blue is:-
$$= \frac{P(B)}{P(w)} = \frac{0.0099}{0.0297} = 0.33$$
- The probability of car actually being blue is 33%.